TRoss SEQ

16-STEP MIDI DRUM SEQUENCER · USER MANUAL



Version 0.9.3 · AU / VST3

Ticco Ross · 2026

Contents

1 What TRoss Seq does
2 Installing
3 Setting up in Live, Logic and Bitwig
4 The panel at a glance
5 Getting the notes to your drum machine
6 Tracks: names, notes, channels
7 Mute, solo and stutter
8 Programming patterns
9 Recording from a keyboard or pads
10 The 12 pattern slots
11 Feel: velocity, accent, gate, swing, flam
12 The randomiser
13 Exporting MIDI
14 Saving and defaults
15 Reference
16 Troubleshooting

1 What TRoss Seq does

TRoss Seq is a 16-step drum sequencer that plays your drum boxes over MIDI, locked to your DAW's tempo: hardware drum boxes, and software drum boxes, whether they're plug-ins in your DAW or apps running on their own.

- **13 drum lanes plus an accent row**, 16 steps each, programmed on lit step pads.
- **Every lane sends its own MIDI note on its own channel**, so one TRoss Seq can play one drum box or several at once, hardware or software.
- **12 pattern slots**. A new slot always waits for the current pattern to finish, so changes land on the bar. **Chain** them (up to 32 steps), locked to the song position.
- **A drum-aware randomiser** that writes patterns the way each drum is usually played, with styles, density, lane locks, variations, fills and undo.
- **Two ways out**: straight to a MIDI port on your computer, or to your DAW's MIDI routing.
- **Flams, drags and ruffs** on any step (**Shift**-click, right-click to edit), and a **drum map library** to keep one map per drum box, with built-in maps for hardware and software drum boxes.
- Swing, gate, accent velocity, MIDI clock out, MIDI learn, keyboard recording and MIDI file export.

TRoss Seq makes no sound of its own. It is an instrument plug-in that outputs MIDI, which is why it loads on a MIDI or instrument track.

2 Installing

macOS

1. Open **TRossSeq-0.9.3-macOS.pkg** (the manual is in the same download).
2. The installer isn't signed, so the first time macOS blocks it. Open **System Settings → Privacy & Security**, scroll down and click **Open Anyway**, then open the .pkg again.
3. Follow the installer. It puts the Audio Unit in `/Library/Audio/Plug-Ins/Components` and the VST3 in `/Library/Audio/Plug-Ins/VST3`. Click **Customize** to install only one of them.
4. In Ableton Live, open **Settings → Plug-Ins**, check that Audio Units (and/or VST3) are switched on, then hold **Option** and click **Rescan**.

Logic and other Audio Unit hosts find it under **AU Instruments → Ticco Ross**; Live and Bitwig under **VST3 → Ticco Ross**. Section 3 has the setup for each.

Windows

Run **TRossSeq-0.9.3-Windows-Setup.exe**. If SmartScreen appears, click **More info → Run anyway**. The VST3 goes to `C:\Program Files\Common Files\VST3`. Rescan plug-ins in your DAW.

Linux

Unpack the `.tar.gz` and run `./install.sh`. It copies the VST3 to `~/.vst3` and the LV2 to `~/.lv2`, with no root needed. `./uninstall.sh` removes them.

Updating

Install the new version over the old one, then rescan. Your projects keep their patterns and settings. The version number is shown at the top right of the panel.

3 Setting up in Live, Logic and Bitwig

Pick the route for the kind of drum box you're playing, then follow the steps for your DAW.

DRUM BOX	ROUTE	WHY
Hardware	**MIDI OUT PORT** → your MIDI interface or the drum box's USB MIDI	Works the same in every DAW, keeps every lane on its own channel, and has its own tight timing.
Software plug-in	**HOST OUT** → routed by the DAW to the plug-in's track	Stays inside the DAW, nothing extra to set up.
Software app (standalone)	**MIDI OUT PORT** → the app's MIDI input, or a virtual MIDI port	The app runs outside the DAW, so it needs a MIDI port (section 6).

In every DAW: open **PRESETS** and pick the map for your drum box (or set each lane's NOTE and CH, section 6), then press play in the DAW. TRoss Seq follows its tempo, start, stop and loops. Use one route per drum box: with both MIDI OUT PORT and HOST OUT reaching the same box, every note plays twice.

Ableton Live

Use the **VST3** version: **Plug-Ins → VST3 → Ticco Ross → TRoss Seq**. Live can't take MIDI out of AU plug-ins, so the AU version only works with MIDI OUT PORT.

Hardware drum box

1. Create a **MIDI track** and drop TRoss Seq on it.
2. Under **MIDI OUT PORT**, pick your interface or drum box. The line underneath should read **CONNECTED**.

Software drum box (plug-in)

1. Put TRoss Seq (VST3) on a MIDI track. Set MIDI OUT PORT to **NONE** and leave **HOST OUT** on.
2. Put the drum plug-in on a second MIDI track.
3. On that second track, set **MIDI From** to the TRoss Seq track, set the chooser below it to **TRoss Seq**, and set Monitor to **In**.

Live merges all MIDI channels into one when it routes between tracks. That's fine for one drum plug-in; to play several plug-ins on different channels, use one TRoss Seq per plug-in.

To play or record from your keyboard, arm the TRoss Seq track (or set its monitor to **In**).

Logic Pro

Logic uses the **Audio Unit**. On a **Software Instrument** track, click the Instrument slot → **AU Instruments → Ticco Ross → TRoss Seq**. If it doesn't appear, open **Logic Pro → Settings → Plug-in Manager** and rescan it.

Hardware drum box: under **MIDI OUT PORT**, pick your interface or drum box.

Software drum box (plug-in), Logic 11 or later

1. On the TRoss Seq track, set MIDI OUT PORT to **NONE** and leave **HOST OUT** on.
2. Create a second Software Instrument track with the drum plug-in.
3. In that track's **Track Inspector**, set **Internal MIDI In** to **Instrument Output** and pick the TRoss Seq track as the source. Set **MIDI In Channel** to the lanes' channel (or All).
4. Turn on **Input Monitoring** on that track and press play.

Logic 10 and earlier can't route MIDI out of an instrument plug-in. There, send TRoss Seq to a virtual MIDI port (section 6), keep the drum plug-in's track selected, and switch TRoss Seq's MIDI THRU off so the notes don't loop back into it.

Bitwig Studio

Bitwig uses the **VST3** (also on Linux). Drag **TRoss Seq** from the browser's plug-ins (Ticco Ross) onto an **Instrument track**.

Hardware drum box: under **MIDI OUT PORT**, pick your interface or drum box.

Software drum box (plug-in)

1. On the TRoss Seq track, set MIDI OUT PORT to **NONE** and leave **HOST OUT** on.
2. Create a second Instrument track with the drum plug-in.
3. In that track's input (the I/O section), pick the TRoss Seq track as its note source, and set the accepted channel to the lanes' channel (or all). Or add a **Note Receiver** device before the plug-in with the TRoss Seq track as its source.

4 If you hear nothing, arm the track or switch its monitoring on.

4 The panel at a glance



TRoss Seq 0.9.3. The window can be resized from its bottom-right corner.

A Feel

VELOCITY, ACCENT, GATE, SWING, FLAM and FLAM LVL knobs (section 11). Turn a knob to see its value; double-click resets it.

B MIDI output

MIDI OUT PORT selector and status line, HOST OUT, CLOCK OUT, MIDI THRU and PORT DELAY (section 5).

C Drum map

PRESETS: built-in maps and your own saved maps. The line below shows the current map, with *** EDITED** when it has changed (section 6).

D Display

Shows the playing pattern and what's queued next. In a DAW TRoss Seq follows the host's transport and tempo. Under it, **STEPS** and **LOCKS** switch the sequencer zone between the step grid and the parameter locks page (section 8).

E Lanes

Name strips (play, rename, lock), step pads, and per lane NOTE, CH, M (mute), S (solo) and LEARN.

F Patterns & transport

12 pattern buttons, LENGTH, RUN, CLEAR, REC and EXPORT MIDI.

G Randomiser

DENSITY, RANDOM STYLE, and the RANDOM, FILL and UNDO keys (section 12).

I About

Click the **TRoss SEQ** logo for the About window: version, build date, which format and DAW it's running in, a quick reference of all mouse and key shortcuts, the contact address (**ticcoross@gmail.com**; click it to write an e-mail), credits, and an **OPEN THE MANUAL** button that opens this manual. The manual comes inside the plug-in, so it works offline and always matches the version you have. Click anywhere else (or press Esc) to close it.

H HELP

Top right. Switches the tips that pop up when you hover a control on or off. Remembered for every copy of TRoss Seq.

Display messages

DISPLAY	MEANING
`PATTERN 1`	Pattern 1 is playing and is the one you're editing.
`PAT 1 > 2`	Pattern 2 is queued. It takes over when pattern 1 is back at step 1.
`PAT 1 > RND`	A random pattern (or fill / variation) is waiting for step 1.
`CHN 2/4 P1`	A chain is playing: step 2 of 4, pattern 1 (section 10).

5 Getting the notes to your drum machine

There are two outputs. Use one or both, but don't send both to the same drum box, or every note plays twice.

OUTPUT	HOW IT WORKS	USE IT IN
MIDI OUT PORT	TRoss Seq sends straight to the port you pick: a MIDI interface, or a drum box with USB MIDI. It has its own high-priority timing, so notes don't jitter with the DAW's buffer.	Every DAW. The way to go for hardware drum boxes.
HOST OUT	The notes go to the DAW's MIDI output for the plug-in, and you route them from there.	Software drum box plug-ins: Live (VST3 version), Logic 11 or later, Bitwig, Reaper, Cubase, Studio One… (section 3).

MIDI OUT PORT

Click the selector to see the ports connected right now; the list refreshes each time it opens. **NONE** switches the port output off. The status line below tells you what's going on:

STATUS	MEANING
CONNECTED	Notes are going to the port.
NOT CONNECTED…	The project asks for a port that isn't plugged in. Plug it in and pick it again, or pick another port.
COULDN'T OPEN…	Another program holds the port (common on Windows; see section 16).

PORT DELAY

Lines the drum box up with the audio coming out of your DAW, from −250 to +250 ms. The knob sits at 0 (no shift); double-click it to go back there.

- **Positive (turn right):** everything sent to the port goes out later. Use it when the drum box plays *ahead* of the DAW's audio, which is the usual case, because the DAW's own sound is delayed by its audio buffer. Start at about your buffer size (in ms).
- **Negative (turn left):** everything sent to the port goes out earlier. Use it when the drum box is *late*, for example a slow USB or MIDI interface, a drum box that reacts slowly, or a software drum box whose audio comes back through a long chain. TRoss Seq runs that much ahead of the song for the port, so the hits still land on the beat.

Adjust by ear against a sound in the DAW. PORT DELAY only moves the MIDI OUT PORT (notes and MIDI clock). HOST OUT always stays exactly on the grid, and MIDI THRU and clicks on the name strips are never moved.

> With a negative delay, the very first step after pressing play can't be sent before you pressed play, so it goes out right away. Over loops, every step is still sent once, and only once.

CLOCK OUT

Sends MIDI clock (24 ticks per quarter note), Start, Stop, Continue and Song Position to the port. Switch it on if your drum box has tempo-synced effects or its own sequencer that should follow along, and set the drum

box to sync to external / MIDI clock. Clock goes to the port only.

MIDI THRU

Passes the MIDI coming into the track (your keyboard or pads) through to the outputs, so you can play the drum box live through TRoss Seq. In Live, arm the track (or set its monitor to **In**) so your keyboard reaches the plug-in.

6 Tracks: names, notes, channels

Each of the 13 lanes has a name strip on the left and its MIDI settings on the right.

CONTROL	WHAT IT DOES
Name strip	**Click** to play the lane's note now; the strip flashes for your click. **Cmd**-click (Mac) or **Ctrl**-click (Windows) plays it at accent velocity. The strip doesn't flash when the sequencer plays the lane.
Rename	**Shift**-click the name and type up to 10 characters; **Return** keeps it, **Esc** cancels. The name also tells the randomiser what kind of drum the lane is (section 12).
NOTE	The MIDI note the lane sends. Drag the red read-out up or down. Note names follow Ableton's convention (C1 = 36).
CH	The MIDI channel, 1–16. Drag up or down.
LEARN	Click it (it blinks), then hit a key or pad: the lane takes that note and channel.
Right-click a name	Rename, Learn, the clear options (section 8), Randomise this lane, Lock from the randomiser.

Drum maps and the PRESETS menu

A **drum map** is the set of lane names, notes, channels and locks. Make one for each drum box you own, save it once, and pick it whenever you use that box. The line under PRESETS shows the current map, e.g. **MAP: TR-8S STUDIO**, with an orange * EDITED once you change anything after picking or saving it.

PRESETS MENU	WHAT IT DOES
Built-in maps	Sets all 13 lanes at once. **General MIDI** drums; **hardware:** Behringer RD-6, RD-8, RD-9; Elektron Digitakt II; Korg volca beats; Roland TR-06, TR-08, TR-09, TR-6S, TR-8 / TR-8S; **software:** Ableton Drum Rack, UVI Prime 8, AudioRealism ADM (ADM1X). The menu groups them by brand. Full tables in section 15.
My maps	Your saved maps. Click one to load it. A tick shows the current, unedited map.
Save "*name*"	Overwrites the current saved map with your changes (only when it's one of your maps and it's edited).
Save map as...	Saves the current lanes under a new name. If the name exists, you're asked before it's replaced.
Rename map / Delete map	Pick a map from the list. Deleted maps go to the bin / trash.
Show maps folder	Opens `Documents/TRoss Seq/Maps`. Each map is one `.tsmap` file: copy them to back up, or to use on another computer.
All lanes on channel...	Moves every lane to one channel.
Save map + MIDI port as default	New copies of TRoss Seq start with the current map and port.

> **Check your drum box.** The preset maps follow each maker's published default notes. If a sound doesn't trigger, look up its note in the drum box's manual or MIDI settings and fix that lane with LEARN or the NOTE read-out.

Playing a software drum box

Plug-ins running in your DAW: see section 3 for Live, Logic and Bitwig.

Standalone apps (a software drum box running on its own, outside the DAW): pick the app's MIDI input under **MIDI OUT PORT** if it creates one. If it doesn't, use a virtual MIDI port and set the app to listen to it. TRoss Seq still follows your DAW's tempo. To make a virtual port on a Mac:

1. Open **Audio MIDI Setup** → Window → Show MIDI Studio, double-click **IAC Driver** and tick **Device is online**.
2. In TRoss Seq, set **MIDI OUT PORT** to **IAC Driver Bus 1** and pick the software's map in PRESETS.
3. In the app, choose **IAC Driver Bus 1** as its MIDI input.
4. Make sure the app listens on the channel the lanes send on (CH), or on all channels.

On Windows, use a virtual MIDI port such as loopMIDI the same way. If your DAW also listens to that virtual port, switch it off as an input there, so the notes don't loop back into TRoss Seq.

Using several drum boxes

Give each drum box its own MIDI channel and set the lanes' CH to match: for example lanes 1–8 on channel 10 for one box, lanes 9–13 on channel 11 for another. Both boxes can sit on the same MIDI port (daisy-chained through MIDI THRU) or you can use two TRoss Seq instances with a port each.

7 Mute, solo and stutter

ACTION	RESULT
Click **M**	Mutes the lane (the button lights yellow). Click again to unmute.
Click **S**	**Solo.** The button lights blue. As soon as any S is lit, only the soloed lanes play, even if they are muted; every other lane goes quiet. Light several S buttons to hear those lanes together. Click again to switch the solo off; when no S is lit, the mutes count again.
Shift + hold **M**	**Momentary solo.** The button turns blue and shows **S**; only that lane plays, even if it is muted itself, and every other lane goes quiet. Let go of the mouse button and everything is back exactly as it was. Nothing is toggled. It beats the S buttons while you hold it.

Mutes and solos can be automated in your DAW (*Track 1 Mute* … *Track 13 Mute*, *Track 1 Solo* … *Track 13 Solo*) and are saved with the project. They don't affect clicks on the name strips or MIDI THRU.

STUTTER (beat repeat)

A live effect at the top of the panel, next to the TRoss SEQ name. **Hold STUTTER** and the slice of the pattern you're on plays over and over. **Let go** and TRoss Seq is straight back where the song is: the song keeps running underneath, so you never fall out of time with the DAW.

CONTROL	WHAT IT DOES
STUTTER	Hold with the mouse: the effect is on while you hold it (the LED lights). Nothing in the pattern changes.

Length (1/2 BAR ... 1/32)	How long the repeated slice is. The slice starts on the grid at the moment you press: with 1/8, pressing on step 5 repeats steps 5 and 6. 1/32 repeats half a step, for machine-gun rolls.
ROLL	The slice gets shorter every two repeats (for example 1/4, 1/4, 1/8, 1/8, 1/16, 1/16, 1/32 ...): the classic build-up into a drop. Let go on the drop.

> **Play it from a controller.** STUTTER, Stutter Length and Stutter Roll are DAW parameters. Map a key or pad of your MIDI controller to *Stutter* in your DAW (in Live: MIDI Map mode; in Logic: Learn in Smart Controls; in Bitwig: right-click the parameter → Learn controller assignment), or draw it as automation to write stutters into an arrangement.

Mutes and solos work during a stutter. Flams are left out of the repeats, so fast stutters stay clean. MIDI clock keeps running at the song's pace, so a drum box following the clock stays in time.

8 Programming patterns

- **Click a pad** to switch a step on or off. **Drag along a lane** to paint several steps on (or off, if you started on a lit pad).
- The pads are coloured in groups of four, so the beats are easy to see. The LEDs above follow the playing step, and the playing column gets a white outline.
- **ACCENT row:** on any step where it's lit, every lane plays at the ACCENT velocity instead of the VELOCITY.

Flams, drags and ruffs

A flam is a quick, softer grace note just before the main hit, like a drummer's two sticks landing almost together. A **drag** has two grace notes and a **ruff** three.

- **Shift**-click a pad to add a flam. If the step was off, it switches on with the flam. **Shift**-click again to remove it; the step stays on. A plain click switches the step off, flam included.
- Grace notes play **before** the step, so the main hit stays exactly on the grid. The pad shows a small column of tiny LEDs on its left edge, one per grace note (1 flam, 2 drag, 3 ruff), and the main LED moves a little to the right. A thin line under the main LED means the step has its own spacing or level.
- The **FLAM** knob sets how far before the step the grace notes start, and **FLAM LVL** how loud they are (section 11). Every flam follows these knobs unless you give it its own settings.

Right-click a pad to edit its flam:

MENU ITEM	WHAT IT DOES
Step on / off	Switches the step.
No flam · Flam · Drag · Ruff	How many grace notes: none, 1, 2 or 3. Picking one on an empty or plain step adds it.
Spacing	**FLAM knob**, or a fixed 12, 25, 40 or 60 ms for this step. The grace notes of a drag or ruff are spread evenly over that time.
Grace level	**FLAM LEVEL knob**, or a fixed 30, 50, 75 or 100 % of the step's velocity for this step.
Use this flam on every hit in this lane	Copies this step's flam type, spacing and level to every step of the lane in the playing pattern.
Remove all flams in this lane	Removes the lane's flams in the playing pattern; the steps stay.

Flams work on every lane except ACCENT, are saved with the project, can be undone with UNDO (whole-lane changes) and are included in EXPORT MIDI. Flams are only ever set by you: the randomiser never adds them (section 12). Removing a flam forgets its settings, so adding it again starts as a plain flam.

Clearing

WHERE	WHAT IT CLEARS
CLEAR	The whole playing pattern, accent row included.
Right-click a name → Clear this lane	That lane's steps (and flams) in the playing pattern.
Right-click a name → Clear this lane's flams	Only the flams; the steps stay.

... in all 12 patterns	The same, in every pattern slot at once.
Right-click the ACCENT label	Clear the accent row, in the playing pattern or in all 12.
Drag along a lane	Starting on a lit pad switches the steps you drag over off.

Every clear can be taken back with UNDO, and one UNDO restores a lane cleared in all 12 patterns.

CONTROL	WHAT IT DOES
LENGTH	Pattern length, 1–16 steps. Drag the red read-out up or down; double-click for 16. Steps past the length are dimmed and don't play. Use odd lengths for patterns that drift against the bar.
RUN	Sequencer on/off. With RUN off, the steps still advance but send nothing; MIDI THRU and the name strips keep working.
CLEAR	Clears the pattern that's playing, accent row included. UNDO brings it back.

TRoss Seq always follows the DAW: it starts, stops, loops and jumps with the transport, and step 1 falls on the downbeat of each bar. When the transport stops, any note still held gets its note-off.

Parameter locks (the LOCKS page)

A parameter lock gives **one step of one lane its own value**, instead of the knob's. Click LOCKS under the display (top right) and the sequencer zone shows the lock page; STEPS brings the step grid back. Both pages show the playing pattern.



The LOCKS page: lane CH, VELOCITY locks on every other step.

1. **Pick a lane** with the names on the left. A lit dot next to a name means that lane has locks in this pattern.
2. **Pick a parameter** on the right.
3. **Set values with the 16 tall bars**, one per step: drag a bar up or down, or drag across several bars to draw a shape (a hi-hat velocity ramp, say). Hold **Shift** while dragging for fine steps.

A **bright orange** bar is a locked step; its value is in the display under it. A **dark** bar (display "-") has no lock and plays at the knob's value, shown by the dashed line. Each bar's cap has the colour of its group of four pads, as on the step grid. Steps that are switched off are drawn dimmer, but you can lock them in advance. The display on the right shows the lane, step and value you're touching.

PARAMETER	RANGE	WHAT THE LOCK DOES
VELOCITY	1–127	How hard the step plays. Replaces the VELOCITY knob, and the ACCENT velocity on accented steps.
GATE	5–95 %	How long the note is held, as part of a 16th note.
NUDGE	±50 %	Micro-timing: moves the hit up to half a step later (up) or earlier (down). Drawn from the centre line.
CHANCE	0–100 %	The probability that the step plays at all. Rolled again every time the step comes round, so a pattern changes from bar to bar.
RATCHET	1–4	The step plays 2, 3 or 4 fast hits spread over its length (shown as slim bars).
PITCH	±12	Plays a different note: the lane's note plus that many semitones. For tuned toms, or a different sound on drum boxes that put sounds on neighbouring notes.
CC 1 – CC 4	0–127	Sends a MIDI CC value just before the step's note, on the lane's channel. Four CCs, so four different controls of your drum box (tune, decay, cutoff…) can be sequenced at once.

Choosing the CC numbers

Next to each CC button is a small display with its CC number (for example `74 CUTOFF`). The numbers are set **per lane**, because drum boxes usually have a control per drum, and they are saved with the drum map. To change one:

- **Click** the display: a menu with common CCs by name and a **Number** sub-menu with all 128.
- **Drag** it up or down to step through the numbers.
- **Double-click** it to type the number. Empty = not used.

A CC showing `--` isn't used: it sends nothing, and its bars can't be set until you pick a number. Look up the numbers in your drum box's MIDI implementation chart.

ACTION ON THE BARS	RESULT
Drag up / down	Locks the step to that value.
Drag across	Draws locks on every step you pass.
Shift-drag	Fine adjustment, starting from the current value.
Double-click	Removes the lock: the step follows the knob again.
Right-click	Clear this parameter on the lane, all locks on the lane, or all locks in the pattern.

Locks belong to the pattern: they are saved with the project, copied with **Copy to**, cleared by CLEAR and by **Clear this lane**, and taken back by UNDO. RANDOM and FILL leave them alone. During a STUTTER the repeats use velocity, gate, pitch, ratchet, CC and chance locks, but not nudge. EXPORT MIDI writes velocity, gate, pitch, nudge, ratchet and CC locks into the clip; chance can't be written into a clip, so the steps are written as always playing (except those at 0 %).

9 Recording from a keyboard or pads

With REC on (it is by default) and the song playing, every incoming note that matches a lane is written into the nearest step of the playing pattern. Swing is taken into account.

- A note matches a lane by **note and channel** first, then by **note alone**, so a keyboard on channel 1 still records into lanes that send on channel 10.
- Velocity above 100 also sets the accent on that step.
- Clicking a name strip while playing records too (**Cmd** / **Ctrl**-click for an accent).
- While the transport is stopped, nothing is recorded; hits just play.

Switch REC off to play along without writing anything.

10 The 12 pattern slots

- **Click a pattern button** to select it. While the song plays, the new slot waits until the current pattern reaches step 1 (its LED blinks while queued), so changes always land cleanly. While stopped, it switches at once.
- The LED shows the slot that's playing; the light cap shows the selected one. A small dash marks empty slots.
- **Right-click a pattern button** to **Copy to** another slot, or **Clear** it.
- The *Pattern* parameter can be automated, so you can arrange pattern changes in your DAW's timeline (not while a chain is set).

Chaining patterns

A chain plays several patterns one after another, for example a groove three times and a fill once.

- **Shift**-click pattern buttons in the order you want them, up to 32 steps. Shift-clicking the same button again repeats it: **1, 1, 1, 2** plays pattern 1 three times, then pattern 2.
- Each pattern plays its full LENGTH, then the next one starts on step 1. At the end, the chain starts again.
- The chain is **locked to the song position**: bar 1 of the song plays chain step 1, bar 2 step 2, and so on (a "bar" being one pattern LENGTH). Jumping, looping or starting in the middle of the song always lands on the right pattern.
- Chained buttons show a dim LED; the one playing is bright. The display shows `CHN 2/4 P1`: chain step 2 of 4, playing pattern 1.
- Editing the pads changes the pattern that's playing in the chain.
- A plain click on a pattern button ends the chain and goes back to that single pattern.
- Right-click a pattern button for **Add to the chain**, **Remove the last chain step** and **Clear the chain**. UNDO takes back chain changes. The chain is saved with the project, and EXPORT MIDI can write it as one clip.

Fill pattern (TR-808 style)

Like the fill-in on the original TR-808: one of your patterns plays **once** as a fill, then the pattern that was playing comes back on its own.

1. **Shift**-click FILL. Its LED **flashes**: TRoss Seq is waiting for you to pick the fill.
2. Click the pattern button you want as the fill. Its LED blinks and the display shows `PAT 1 > FILL 5`.
3. At the next step 1, the fill plays for one bar (its LENGTH); the display shows `FILL 5`. Then the previous pattern carries on.

The FILL LED stays lit while a fill is waiting or playing. **Shift**-click FILL again to cancel. With a chain running, the fill takes the place of one bar and the chain then carries on from the song position. Stopping the song in the middle of a fill goes back to the previous pattern.

11 Feel: velocity, accent, gate, swing, flam

KNOB	RANGE	WHAT IT DOES
VELOCITY	1–127	Velocity of normal steps. Default 100.

ACCENT	1–127	Velocity of steps with the ACCENT row lit, and of **Cmd** / **Ctrl**-clicked name strips. Default 127.
GATE	5–95 %	How long each note is held, as a fraction of a 16th note. Most drum boxes only care about the note-on; longer gates matter for sounds that respond to note length.
SWING	50–75 %	50 % is straight. Higher values push every second 16th (steps 2, 4, 6…) later: 54–58 % is a light shuffle, 66.7 % a triplet feel, 75 % the maximum.
FLAM	5–80 ms	How far before a flammed step its grace notes start. Default 25 ms; 10–20 ms is tight, 40 ms and up sounds like separate hits. Never more than half a step. Steps with their own spacing (right-click a pad) ignore it.
FLAM LVL	10–100 %	How loud grace notes are, as a percentage of the step's velocity. Default 60 %. Steps with their own grace level ignore it.

12 The randomiser

The randomiser writes patterns lane by lane, the way each drum is usually played, instead of throwing random dots on the grid.

CONTROL	WHAT IT DOES
RANDOM	Writes a **whole new pattern** into the playing slot. Every unlocked lane is rewritten over the current LENGTH. **The ACCENT row is never changed.**
Shift-click RANDOM	A **variation**: only a few steps of a few lanes change, and the anchors (kick on step 1, the backbeat) stay. Good for keeping a groove alive without losing it.
FILL	Rewrites only the **last four steps** of the pattern with a fill: toms, snare, hats and percussion. Press it before the bar you want the fill in. **Shift**-click FILL for the TR-808 style fill pattern instead (section 10).
UNDO	Goes back to the pattern as it was before the last RANDOM, variation, fill, lane randomise, CLEAR, lane clear or whole-lane flam change; up to 32 steps back. Greyed out when there's nothing to undo.
DENSITY	How busy random patterns are. 50 % follows the style as designed; lower is sparser, higher is busier. The anchors (kick on 1, the backbeat in STRAIGHT) are kept at any density.
RANDOM STYLE	The character of what RANDOM and the variation write (below).
Alt-click a name (**Option**-click on a Mac)	**Locks** the lane: a padlock appears and RANDOM, the variation and FILL leave it alone. Alt-click again to unlock. Locks are saved with the project.
Right-click a name → Randomise this lane	Rolls just that one lane, even if it's locked.

Timing

While the song plays, a random result **waits for step 1**, so the pattern never changes in the middle of a bar. The RANDOM key's light blinks and the display shows `PAT 1 > RND` until it takes over. Press RANDOM again while it's waiting to roll a different one; press UNDO to cancel it. While stopped, results are written at once.

Styles

STYLE	CHARACTER
STRAIGHT	Four on the floor: kick on the beats with the odd extra, snare always on the backbeat (steps 5 and 13), 8th-note hats with some 16ths, open hats on the off-beats, toms near the end of the bar.
ELECTRO	Syncopated kick (always on step 1, then pushed around steps 4, 7, 11 and 12), snare on step 13 and usually 5, clap mostly on 13, busy 16th hats, occasional open hats and tom fills.
BROKEN	Scattered and off the grid: kick on 1, then anywhere; snare not tied to the backbeat; looser hats and more percussion.
HOUSE	Kick on every beat, open hat on every off-beat, clap on 5 and 13, shuffled 16th closed hats. Pairs well with SWING.

TECHNO	Driving four-on-the-floor kick, sparse claps, rolling 16th hats, rim and percussion building tension. Darker and busier than HOUSE.
DETROIT	Soulful techno: syncopated hats and claps, cowbell and clave figures, the odd extra kick.
MIAMI BASS	Booming, bouncy kick figures with double hits, snare on 13, fast hats and tom runs.
BREAKBEAT	Funky break: kick on 1 and around 11, snare on 5 and 13 with ghost notes, busy hats with gaps.
HIP-HOP	Laid back: sparse kick, heavy snare on 5 and 13, simple 8th hats. Use it with SWING.
DUB	Very sparse and spacious, with rim shots instead of a snare backbeat and long gaps.
LATIN	Built around the son clave on the claves, rim and cowbell lanes, with toms and agogo interlocking.
INDUSTRIAL	Hard and mechanical: kick on every beat and snare on the backbeat, locked to the grid, few hats.

Flams and the randomiser

The randomiser never adds flams. `RANDOM` and **Randomise this lane** remove the flams on the lanes they rewrite; locked lanes keep theirs, and variations and FILL keep flams on steps that stay on. `UNDO` brings removed flams back.

How lanes are recognised

The randomiser decides what each lane is from its **name** first, so renaming a lane changes how it's treated. If the name doesn't say, it goes by the lane's **General MIDI note**.

TREATED AS	NAMES THAT COUNT	GM NOTES	PLAYS LIKE
Kick	BD, KICK, BASS…	35, 36	Anchored on step 1, plus beats or syncopation
Snare	SD, SNARE	38, 40	Backbeat, ghost notes
Clap	CP, HC, CLAP	39	Backbeat, sometimes
Closed hat	CH, HH, CLOSED, CL HAT, HAT	42, 44	Busy 8ths and 16ths
Open hat	OH, OPEN, OP HAT	46	Off-beats, never on the same step as a closed hat
Cymbal	CY, CC, RD, RC, CRASH, RIDE	49, 51, 52, 53, 55, 57, 59	Rare, mostly on step 1
Tom	TOM, LT, MT, HT	41, 43, 45, 47, 48, 50	Sparse, near the end of the bar
Percussion	RS, RIM, CB, COWBELL, CL, CLAVES, AGOGO, CONGA…	everything else	Sparse off-beat accents

> **Workflow tip.** Program or keep a kick and snare you like, Alt-click to lock them, then press RANDOM until the hats and percussion groove. Shift-click RANDOM for variations, FILL before a change, and UNDO whenever you've gone one roll too far.

13 Exporting MIDI

Click EXPORT MIDI for a menu:

- **Pattern N (playing)** saves the playing pattern as a `.mid` file.
- **All patterns, in a row** saves every non-empty slot one after another in a single file.
- **All patterns as separate files** saves one file per non-empty slot into a folder you choose.
- **The chain, in a row** saves the chain as one file, pattern after pattern (when a chain is set).

Drag the EXPORT MIDI key onto a MIDI track in your DAW to drop the playing pattern in as a clip.

Exported files use each lane's note and channel, the current VELOCITY, ACCENT, GATE, SWING, FLAM, LENGTH and the song tempo, with 16th-note steps. Flams, drags and ruffs are written as grace notes before the step.

14 Saving and defaults

Everything is saved with your DAW project: all 12 patterns with their flams, the playing slot, the drum map (and whether it was edited), every knob and switch, and the MIDI port. If the saved port isn't connected when the project opens, the status line says so, and TRoss Seq reconnects to a port with the same name when it's available.

PRESETS → Save map + MIDI port as default stores your lane names, notes, channels, locks and port, and every new copy of TRoss Seq starts with them. Saved drum maps live in `Documents/TRoss Seq/Maps` and are shared by all your projects. The HELP setting is remembered for every copy of TRoss Seq.

15 Reference

Built-in maps

Notes are taken from each maker's MIDI implementation chart or manual. Note names: C1 = 36. Spare lanes are sounds the machine doesn't have: they're dimmed, send note 0 if you program them, and the randomiser leaves them alone. Give one a name and note to use it.

Hardware

LANE	ROLAND TR-06 (CH 10)		ROLAND TR-08 (CH 1)		ROLAND TR-09 (CH 10)	
1	BD	36	BD	36	BD	36
2	SD	38	SD	38	SD	38
3	LT	47	LT	43	LT	43
4	HT	50	HT	50	HT	50
5	spare	–	CP	39	HC	39
6	CH	42	CH	42	CH	42
7	OH	46	OH	46	OH	46
8	CY	49	CY	49	CC	49
9	spare	–	RS	37	RS	37
10	spare	–	MT	47	MT	47
11	spare	–	MA	70	RC	51
12	spare	–	CB	56	spare	–
13	spare	–	CL	75	spare	–

LANE	ROLAND TR-8 / TR-8S (CH 10)		KORG VOLCA BEATS (CH 10)	
1	BD	36	KICK	36
2	SD	38	SNARE	38
3	LT	43	LO TOM	43
4	HT	50	HI TOM	50
5	HC	39	CLAP	39
6	CH	42	CL HAT	42
7	OH	46	OP HAT	46
8	CC	49	CRASH	49
9	RS	37	CLAVES	75
10	MT	47	AGOGO	67
11	RC	51	RIM	37
12	CB	56	MID TOM	47
13	CL	75	RIDE	51

TR-08: the three congas (high 62, mid 63, low 64) don't fit in 13 lanes; set any lane to them with NOTE or LEARN. TR-08 receives on channel 1 by default, the others on channel 10; if you changed the channel on the machine, use PRESETS → All lanes on channel.

LANE	BEHRINGER RD-8 (CH 1)		BEHRINGER RD-9 (CH 1)		BEHRINGER RD-6 (CH 1)	
1	BD	36	BD	36	BD	36
2	SD	40	SD	40	SD	40
3	LT	45	LT	45	LT	45
4	HT	50	HT	50	HT	50
5	CP	39	CP	39	CP	39
6	CH	42	CH	42	CH	42
7	OH	46	OH	46	OH	46
8	CY	51	CC	49	CY	51
9	RS	37	RS	37	spare	–
10	MT	47	MT	47	spare	–
11	spare	–	RC	51	spare	–
12	CB	56	spare	–	spare	–
13	spare	–	spare	–	spare	–

LANE	ROLAND TR-6S (CH 10)		ELEKTRON DIGITAKT II (CH = LANE)	
1	BD	36	BD	60
2	SD	38	SD	60
3	LT	43	LT	60
4	spare	–	HT	60
5	HC	39	CP	60
6	CH	42	CH	60
7	OH	46	OH	60
8	spare	–	CY	60
9	spare	–	RS	60
10	spare	–	MT	60
11	spare	–	RD	60
12	spare	–	CB	60
13	spare	–	PERC	60

Behringer RD-8, RD-9 and RD-6: notes from the RD manuals' MAP tables. They receive on channel 1 by default. Roland TR-6S: its six instruments on channel 10. Elektron Digitakt II: each track has its own MIDI channel (track 1 on channel 1, track 2 on channel 2 …), so each lane plays its track on that channel with note 60 (C4), the sample at its own pitch. The names are only a starting point: rename the lanes to match your kit.

General MIDI and software

LANE	GENERAL MIDI DRUMS (CH 10)		UVI PRIME 8 (CH 1)		AUDIOREALISM ADM (ADM1X) (CH 1)	
1	BD	36	BD	36	BD	36
2	SD	38	SD	38	SD	40
3	LT	45	LT	41	LT	43

4	HT	50	HT	45	HT	50
5	CP	39	CP	39	CP	39
6	CH	42	CH	42	CH	44
7	OH	46	OH	46	OH	46
8	CY	49	CY	49	CY	51
9	RS	37	RS	37	RS	37
10	MT	47	MT	43	MT	47
11	RD	51	spare	–	spare	–
12	CB	56	CB	47	CB	49
13	CL	75	spare	–	PCF	52

UVI Prime 8: notes from the Prime 8+ manual. Its pitched bass drum (C-1 to B0) and pattern keys (C3 to C4) aren't used.

LANE	ABLETON DRUM RACK (CH 1)		
1	BD	36	C1
2	SD	38	D1
3	LT	41	F1
4	HT	48	C2
5	CP	39	D#1
6	CH	42	F#1
7	OH	46	A#1
8	CY	49	C#2
9	RS	37	C#1
10	MT	45	A1
11	RD	51	D#2
12	TOM	43	G1
13	PD HAT	44	G#1

Ableton Drum Rack: the 16 pads of a Drum Rack sit on C1 to D#2. The map follows the General MIDI layout most Ableton kits use. Pads E1 (40) and B1 (47) are left over: set a lane to them with NOTE or LEARN if your kit has a sound there. Each Ableton kit puts its own sounds on the pads, so rename lanes (Shift-click) to match yours.

AudioRealism ADM: put ADM in **Note mode** (its own sequencer off). Notes from the ADM manual's note-mode table, using the unaccented keys; TRoss Seq's accented steps (velocity above 100) play ADM's accents. Lane 13 triggers ADM's **PCF** (Pattern Controlled Fx); the randomiser leaves it alone.

Mouse and keyboard

WHERE	ACTION
Step pad	Click: on/off · Drag: paint
	Shift-click: flam on / off
	Right-click: flam type / spacing / level

WHERE	ACTION
RANDOM	Click: new · Shift: variation

Name strip	Click: play · Cmd/Ctrl: accent
	Shift-click: rename
	Alt / Option-click: lock
	Right-click: clear lane, learn…
ACCENT label	Right-click: clear accent row
STEPS / LOCKS	Step grid / parameter locks page
FILL	Click: random fill · Shift-click, then a pattern: TR-808 fill (plays once)
STUTTER	Hold: beat repeat · length 1/2 bar … 1/32 · ROLL: shorter every two repeats
M	Click: mute · Shift + hold: solo while held
S	Click: solo (several at once)

Pattern button	Click: select · Right-click: copy / clear / chain
	Shift-click: add to the chain
Knob	Drag · Double-click: default
Red read-outs	Drag up / down
EXPORT MIDI	Click: menu · Drag: clip
HELP	Hover tips on / off
TRoss SEQ logo	Click: About + quick reference

Automatable parameters

Run · Length · Pattern · Record · Velocity · Accent Velocity · Gate · Swing · Flam Time · Flam Level · Port Delay · MIDI Thru · Host Out · Clock Out · Random Density · Random Style · Track 1–13 Mute · Track 1–13 Solo · Stutter · Stutter Length · Stutter Roll

16 Troubleshooting

PROBLEM	WHAT TO CHECK
The drum box plays nothing	Is the song playing and RUN on? Does the MIDI OUT PORT status read CONNECTED? Do the lanes' CH match the channel the drum box listens on (often 10)? Are the notes right for your drum box (PRESETS, LEARN)? Is the lane muted, or is another lane soloed (S lit)? For a software plug-in, check the routing in section 3 for your DAW.
Every note plays twice	Both outputs reach the same drum box. Turn HOST OUT off, or set MIDI OUT PORT to NONE.
One sound doesn't trigger	That lane's note isn't what the drum box expects. Click LEARN on the lane and hit the pad on the drum box (if it sends MIDI), or look up the note in its manual.
Port shows COULDN'T OPEN (Windows)	Most Windows MIDI drivers let only one program use a port. Switch that port off in your DAW's MIDI settings, close other MIDI apps, then pick it again in TRoss Seq.
Port shows NOT CONNECTED	The port saved in the project isn't plugged in. Connect it and pick it again, or choose another.
Drum box is early or late against the DAW	Adjust PORT DELAY until they line up: turn it right if the drum box is early, left (negative) if it's late.
Drum box's tempo / effects don't follow	Switch CLOCK OUT on and set the drum box to sync to external MIDI clock.
RANDOM doesn't change anything at once	While playing, the result waits for step 1; the display shows `PAT 1 > RND`. Check the lane isn't locked (padlock).
Keyboard doesn't play or record	Arm the TRoss Seq track (in Live you can also set its monitor to In). MIDI THRU must be on to pass it to the drum box; REC must be on and the song playing to record.
The pop-up tips get in the way	Switch **HELP** off (top right). Knob values still show while you turn them.
A flam doesn't sound like a flam	Some drum boxes cut a sound short when it retriggers quickly, or ignore low velocities. Try a longer FLAM time, a higher FLAM LVL, or raise VELOCITY.
TRoss Seq doesn't appear in the DAW	Rescan plug-ins (Live: Settings → Plug-Ins, Option-click Rescan on a Mac; Logic: Plug-in Manager; Bitwig: Settings → Locations, rescan). In Live check Audio Units / VST3 are switched on. On a Mac, check the installer actually ran (Privacy & Security → Open Anyway).